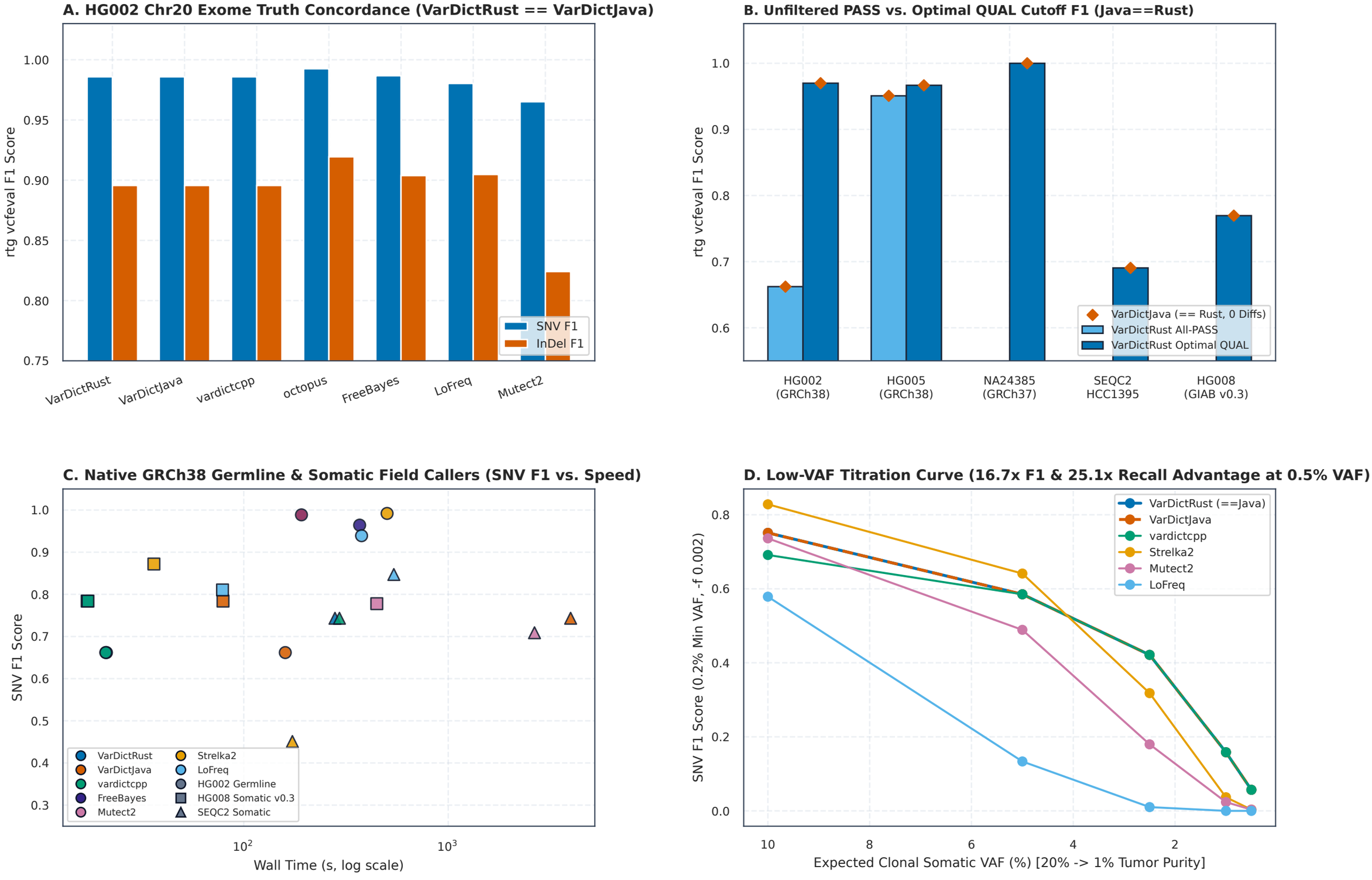
A. HG002 Chr20 Exome Truth Concordance (VarDictRust == VarDictJava)
rtg vcfeval F1 Score
1.00
0.95
0.90
0.85
0.80
0.75
SNV F1
InDel F1
VarDictRust
VarDictJava
vardictcpp
octopus
FreeBayes
LoFreq
Mutect2
B. Unfiltered PASS vs. Optimal QUAL Cutoff F1 (Java==Rust)
rtg vcfeval F1 Score
1.0
0.9
0.8
0.7
0.6
VarDictJava (== Rust, 0 Diffs)
VarDictRust All-PASS
VarDictRust Optimal QUAL
HG002
(GRCh38)
HG005
(GRCh38)
NA24385
(GRCh37)
SEQC2
HCC1395
HG008
(GIAB v0.3)
C. Native GRCh38 Germline & Somatic Field Callers (SNV F1 vs. Speed)
SNV F1 Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
VarDictRust
VarDictJava
vardictcpp
FreeBayes
Mutect2
Strelka2
LoFreq
HG002 Germline
HG008 Somatic v0.3
SEQC2 Somatic
10^2
10^3
Wall Time (s, log scale)
D. Low-VAF Titration Curve (16.7x F1 & 25.1x Recall Advantage at 0.5% VAF)
SNV F1 Score (0.2% Min VAF, -f 0.002)
0.8
0.6
0.4
0.2
0.0
VarDictRust (==Java)
VarDictJava
vardictcpp
Strelka2
Mutect2
LoFreq
10
8
6
4
2
Expected Clonal Somatic VAF (%) [20% -> 1% Tumor Purity]